AETHRA-SEC

Security Operations Console

Installation and User Manual

Version 1.0.0

A Unified System for Automated Penetration Testing with Real-Time Network Traffic Analysis and Immediate Threat Mitigation Recommender

IMPORTANT
AETHRA-SEC must only be operated against networks, hosts and interfaces you are explicitly authorised to monitor or assess. It performs no exploitation, no credential attacks and no automated mitigation. All mitigation output is advisory and requires a human decision.

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1. About this manual

This manual describes AETHRA-SEC version 1.0.0 as implemented in this repository. It was written by reading the application itself: the installer script, the setup wizard, the settings registry, the health service and the screen definitions. Screenshots are captures of the running application against a live database.

What AETHRA-SEC is

A Windows desktop security operations console. It captures network traffic, ingests intrusion detection alerts from a Snort sensor, runs authorised Nmap assessments, maintains a host inventory, and recommends mitigations from a local knowledge base. It records what it observes and never fabricates an observation.

What AETHRA-SEC does not do

- It does not exploit hosts, attack credentials, or apply mitigations.
- It does not decrypt HTTPS. Only metadata the capture genuinely carries is displayed.
- It does not launch Snort. It reads alerts a sensor has written.
- It cannot see traffic the network does not deliver to its capture interface. Section 10 explains this in detail.

Audience	Start at
Installing for the first time	Sections 2 to 12
Operating a working installation	Sections 13 to 26
Something is wrong	Section 29, Troubleshooting
Upgrading or removing	Sections 27 and 28

2. Before you install

AETHRA-SEC ships as a Windows installer that contains its own Python runtime and every Python library it needs. You do not need Python, an IDE, or any development tool to run the installed application.

What you do need is the security tooling AETHRA-SEC integrates with. Those are separate products with their own licences and are installed separately. The installer reports which are already present; it does not install them for you.

Requirements

Requirement	Detail
Operating system	Windows 10 or 11, 64-bit
Privileges	Administrator rights to install. Live packet capture requires an elevated process; the installed shortcut prompts for elevation.
Disk space	Approximately 24 MB for the installer payload, plus room for logs, captures and reports under the data directory. Capture files are bounded by a ring buffer (Section 25).
Database	MySQL 8.0+, MariaDB 10.6+, or XAMPP, running, with an account permitted to create a schema.
Network	Only to reach the database and the hosts you are authorised to assess. AETHRA-SEC does not phone home.

IMPORTANT
Being connected to the same LAN does not let a Windows laptop see every other device's traffic. If you intend to monitor devices other than this machine, read Section 10 before installing — it may change where you install AETHRA-SEC.

3. Third-party dependencies

AETHRA-SEC detects each dependency at startup and re-checks it every time you open System Health. A missing dependency never prevents the application from starting: the affected screen reports the component as unavailable rather than implying it is working.

Component	Required	Used for	If missing
MySQL / MariaDB	Yes	All storage: hosts, alerts, findings, audit trail	The application starts with the database reported OFFLINE; nothing can be stored
Npcap	Yes for capture	The Windows packet driver	Live capture is impossible
Wireshark / TShark	Yes for capture	Traffic analysis, driven through PyShark	Live Monitoring reports TSHARK UNAVAILABLE
Nmap	Yes for ass essment	Host discovery and vulnerability assessment	Assessment screens report NMAP UNAVAILABLE
Snort	Optional	Intrusion detection alerts	AETHRA-SEC's own traffic rules continue; interface reports SNORT UNAVAILABLE

Installing them

- **Npcap** is included in the Wireshark installer. Keep that component selected; installing Wireshark covers both Npcap and TShark.
- **PyShark** is a Python library and is already inside the AETHRA-SEC installer. You do not install it separately.
- **Nmap** and **Snort** are installed from their own distributions.

How AETHRA-SEC finds them

Each tool is located on PATH or at its standard install location. You can override any path in the configuration file using the variables shown in Section 6. The setup wizard's *Security tools* step reports what it found, with the executable path and version it actually ran.

Verifying by hand

```
nmap --version
tshark --version
snort -V
mysql --version
```

4. Installing AETHRA-SEC

AETHRA-SEC is installed with **AETHRA-SEC-Setup.exe**, an Inno Setup installer. The installer places the application and its runtime; it does not configure the application. Configuration happens at first launch, in the setup wizard described in Section 5.

Step	What happens	What you do
1. Elevation	Windows asks for administrator rights.	Accept. Installation writes to Program Files.
2. Existing install check	If AETHRA-SEC is already installed the installer says so and offers to continue.	Continue to upgrade in place, or cancel.
3. Licence notice	Third-party licence information is shown.	Read and accept.
4. Prerequisites	The installer lists the security tooling and reports what is already on the machine.	Note anything missing. It can be installed later.
5. Destination	Install location, default `C:\Program Files\AETHRA-SEC`.	Accept or choose another directory.
6. Shortcuts	Optional Start menu and desktop icons.	Select what you want.
7. Install	Files are copied and the data directory is created with modify permission for users.	Wait.
8. Finish	Offers to start AETHRA-SEC immediately.	Leave ticked to go straight to first-run setup.

Shortcuts the installer creates

Shortcut	Command	Purpose
AETHRA-SEC	AETHRA-SEC.exe	The console
AETHRA-SEC diagnostics	aethra-sec-cli.exe --check	Runs the health check without opening the interface
AETHRA-SEC setup	AETHRA-SEC.exe --setup	Re-runs the setup wizard
Uninstall AETHRA-SEC	unins000.exe	Removal (Section 28)

Installing from source instead

If you are working from a source checkout rather than the installer, run `setup.bat`. It prepares the Python environment and then hands over to the same setup code the graphical wizard runs — the same dependency probes, configuration writer, database provisioning and administrator creation. There is deliberately only one implementation. It is safe to run again.

5. First-run setup wizard

The first time AETHRA-SEC starts it runs a seven-step wizard. Nothing is written until the *System health check* step applies it, so you can move back and forth freely. You can re-run it later from the *AETHRA-SEC setup* shortcut.

Step	What it asks for	Notes
1. Administrator account	Username, password, confirmation, email	Creates the first administrator. The password is written into a hash and never stored in readable form.
2. Database	Host, port, schema name, user, password	The wizard creates the schema and applies all migrations.
3. Security tools	Paths to Nmap, TShark and Snort	Leave blank to auto-detect. The step reports what it found and the version it ran.
4. Network interface	The capture interface	Lists real adapters. Read Section 10 before choosing.
5. Email notifications	SMTP host, port, credentials, sender	Optional. Leave blank to skip; notifications remain visible in the application.
6. System health check	Nothing — it applies and verifies	Writes the configuration, provisions the database, creates the administrator, then re-probes every component.
7. Finish	Nothing	Summary. Close to open the sign-in window.

> **NOTE**
> If a step fails, the wizard reports the actual error from the tool or database rather than a generic message. Fix the underlying cause and use Back and Next to retry — nothing partial is left behind.

6. Installation layout

Application files and operator data are deliberately separated. The application directory is replaced on upgrade; the data directory is not.

Location	Contents
C:\Program Files\AETHRA-SEC	The executable, the bundled Python runtime, libraries and assets. Replaced on upgrade.
C:\ProgramData\AETHRA-SEC	The data directory, created by the installer with modify permission for users.
\logs	Application log and interface state.
\captures	Packet capture files, bounded by the ring buffer.
\reports_output	Generated reports.
configuration file	Database and SMTP settings. Tightened by the application when setup writes it, because it holds passwords.
MySQL / MariaDB	All security records: hosts, alerts, incidents, findings, audit trail. Outside both directories above.

IMPORTANT
The configuration file contains the database password and, if configured, the SMTP password. Do not copy it into a report, an issue tracker or a version control system.

7. Database setup

AETHRA-SEC stores every security record in MySQL or MariaDB. The setup wizard provisions this for you: it creates the schema if it does not exist and applies every migration in order. You do not need to create tables by hand.

Applying the schema manually

If you prefer to provision the schema yourself, the repository ships a generated SQL file. It contains the reference rows a working installation needs — roles, permissions, settings and the mitigation knowledge base — and no user accounts and no security records.

```
mysql -u root -p < sql\aethra_sec_schema.sql
python scripts\init_db.py
```

Migrations

The schema is versioned by forward-only migrations recorded in the `schema_migrations` table. On startup AETHRA-SEC applies any migration newer than the recorded version. Start with `--no-migrate` to skip this, for example when running under a least-privilege account that cannot alter tables.

IMPORTANT
Never drop or recreate the database as part of a normal installation or upgrade. Migrations are additive and preserve your existing hosts, alerts, findings and audit trail.

Least-privilege account

After the first run the application needs only SELECT, INSERT, UPDATE and DELETE. CREATE, ALTER, INDEX, REFERENCES and DROP are needed only while migrations run. The shipped SQL file documents a suitable grant.

8. Administrator account

The first administrator is created by the setup wizard. Further accounts are created inside the application from the Users screen (Section 23).

Field	Meaning
Username	The sign-in name. Must be unique.
Password	Shown in this manual only as <administrator password>. Stored as an Argon2id hash; the plain value is never written anywhere.
Full name	Optional display name.
Email	Optional. Used for notifications when SMTP is configured.
Role	ADMIN or USER. Roles carry named permissions which every service re-checks independently of the interface.
Status	Active or inactive. The last active administrator cannot be deactivated or demoted.

NOTE
Two-factor authentication is not part of the current implementation and is not described in this manual.

9. Nmap configuration

Nmap performs host discovery and vulnerability assessment. AETHRA-SEC locates the executable on PATH or at its standard install location, and the setup wizard reports the path and version it actually ran.

Authorisation boundary

AETHRA-SEC refuses to scan outside a configured set of networks. The boundary is enforced in the service layer, not in the interface, so it cannot be bypassed by the screen. It defaults to private ranges and loopback; scanning public addressing requires an explicit, deliberate change.

```
AETHRA_AUTHORIZED_NETWORKS=192.168.0.0/16,10.0.0.0/8,172.16.0.0/12,127.0.0.0/8
AETHRA_ALLOW_PUBLIC_TARGETS=false
```

IMPORTANT
Only assess hosts you are authorised to assess. Widening the boundary does not confer authorisation.

10. Network interface and capture visibility

This is the section most likely to explain an unexpected result, and it is about your network rather than about AETHRA-SEC.

Choosing an interface

AETHRA-SEC enumerates real adapters and presents the interface associated with the active default route first, ahead of virtual, disconnected and loopback adapters. It never silently selects an arbitrary adapter, and you can always choose manually. The selected adapter, its description and its address are shown on the Live Monitoring screen.

What a capture interface can and cannot see

AETHRA-SEC displays only what Npcap delivers to the selected interface. Two mechanisms are commonly confused:

- **Promiscuous mode** lets an adapter keep frames that reach it but are addressed to some other device.
- **Mirroring (SPAN) or a network TAP** is what causes another device's frames to reach the interface at all.

Promiscuous mode alone therefore cannot make a switched or wireless network deliver another device's unicast traffic. AETHRA-SEC always requests promiscuous mode; whether the adapter grants it is a property of the driver.

> **IMPORTANT**
> On Wi-Fi, a station is not delivered other stations' unicast frames, and under WPA2/WPA3 each station's traffic is encrypted with its own pairwise key. Many Windows Wi-Fi adapters also report promiscuous mode as unsupported outright. To monitor other devices you need an authorised observation point: switch SPAN or port mirroring, a network TAP, or a capture point on the gateway or router.

Checking what you can actually see

The capture diagnostic reports, from real captured frames rather than from the host inventory, how many local and non-local hosts have been observed and how many peer unicast frames arrived. A host recorded in the Hosts table was seen at some point; that is not evidence that its traffic is visible now. Windows reports adapter capability directly:

```
netsh wlan show wirelesscapabilities
```


11. Snort configuration

Snort is optional. AETHRA-SEC **reads alerts that a Snort sensor has already written**; it never launches or supervises Snort, and it does not ship or author Snort rules. Detection quality is a property of your rule set.

The alert path

```
network traffic -> Snort -> rule match -> alert output
-> AETHRA-SEC ingest -> MySQL -> IDS Alerts
```

Transports

Configure how AETHRA-SEC receives alerts with `AETHRA_SNORT_TRANSPORT`:

Value	How alerts arrive
file	AETHRA-SEC tails a Snort alert file (alert_fast, alert_csv or alert_json). Recommended on Windows.
unixsock	Snort's alert_unixsock plugin, on Linux or WSL.
tcp	AETHRA-SEC listens on a TCP port for line-oriented alerts.
syslog	AETHRA-SEC listens on a UDP syslog port.
disabled	No ingest. The interface reports Snort unavailable.

> **NOTE**
> With Snort disabled or absent, AETHRA-SEC's own traffic-analysis detection rules continue to raise anomalies. The interface distinguishes the two sources rather than presenting them as one.

12. First launch

A working first session, in order:

#	Action	What to confirm
1	Start AETHRA-SEC from the shortcut	Accept the elevation prompt. Capture needs it.
2	Complete the setup wizard	Seven steps, Section 5.
3	Sign in	With the administrator account you just created.
4	Open System Health	Database, Nmap, TShark, capture and Snort are probed live.
5	Open Live Monitoring	Confirm the adapter shown is the one you intend to monitor.
6	Start capture	Packets should begin arriving within seconds.
7	Open the Dashboard	Counters reflect what the subsystems actually reported.

Running the health check without the interface

```
aethra-sec-cli.exe --check
```


13. Signing in

STEP 1 — OPEN AETHRA-SEC AND SIGN IN

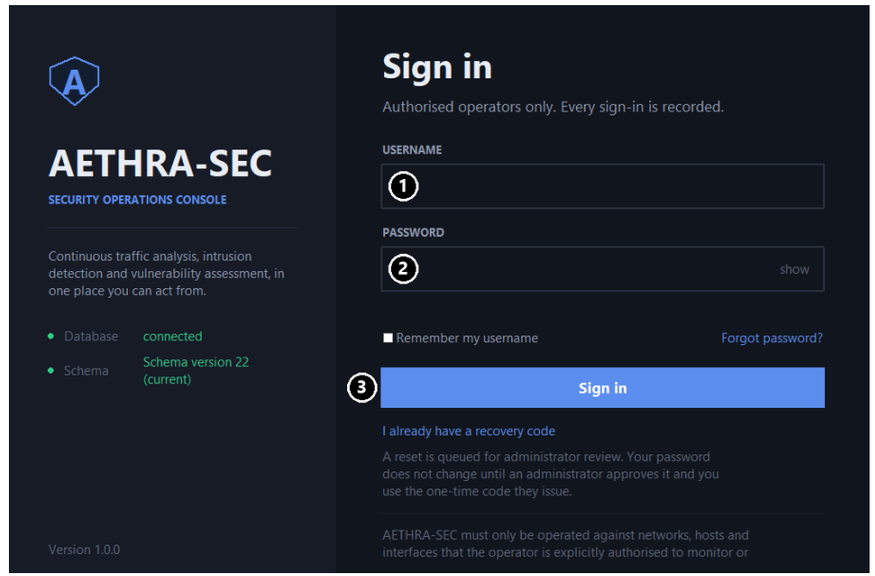


Figure 1. The AETHRA-SEC sign-in window.

①	**Username** Enter your AETHRA-SEC account username.
②	**Password** Enter your account password. It is never shown and never stored in readable form.
③	**Sign in** Select Sign in to authenticate and open the console.

Instructions

1. Start AETHRA-SEC from the desktop or Start menu shortcut and accept the elevation prompt. Live capture needs an elevated process.
2. Enter your account username.
3. Enter your password. Select *show* if you need to check what you typed.
4. Select **Sign in**. The console opens on the screen you last used, if your role still permits it.

Topic	Behaviour
Sign in	Username and password. The console opens on the screen you last used, if your role still permits it.

Topic	Behaviour
Failed attempts	Every attempt is recorded, successful or not, with the host it came from.
Lockout	Repeated failures lock the account. An administrator unlocks it from the Users screen.
Sessions	Active sessions are listed on the Users screen and can be ended by an administrator.
Sign out	Returns to the sign-in window without restarting the application or the background services.

NOTE
Password recovery requests raised at the sign-in screen appear in a queue on the Users screen. Nothing changes until an administrator decides.

14. Dashboard

The landing screen: live security posture assembled from real subsystem data. Figures come from stored records and live counters, never from estimates.

STEP 2 — READ THE SECURITY POSTURE

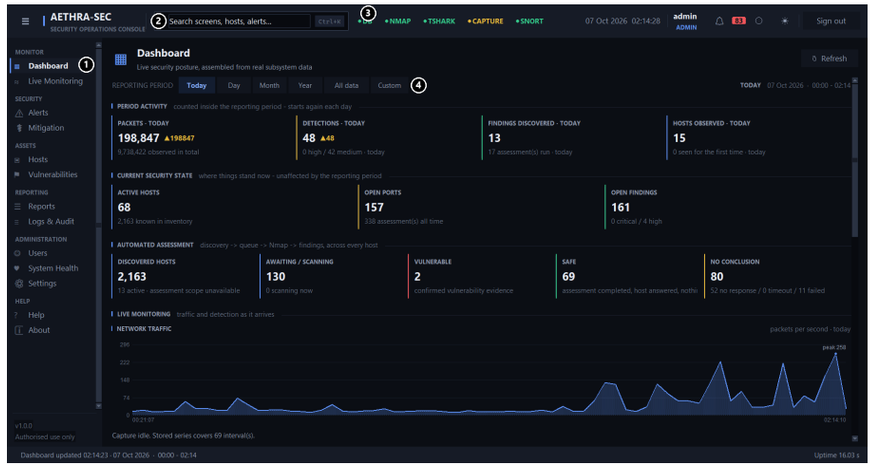


Figure 2. The Dashboard, showing real counters.

①	**Navigation sidebar** Every screen your role permits, grouped by the question it answers.
②	**Command bar** Search screens, hosts and alerts. Ctrl+K opens it from anywhere.
③	**Subsystem status** Live state of the database, Nmap, TShark, capture and Snort.
④	**Reporting period** Changes which records the period counters cover. It does not affect the current-state band.

Instructions

1. Choose a reporting period. It changes the period counters only; the current-state band is unaffected.
2. Read the current security state for active hosts, open ports and open findings as they stand now.
3. Check the subsystem indicators in the header. Any one of them not green is explained on the System Health screen.

Band	What it shows
Reporting period	Today, Day, Month, Year, All data or Custom. Changes which records the period counters cover.
Period activity	Packets, detections and findings inside the selected period.

Band	What it shows
Current security state	Active hosts, open ports and open findings as they stand now, unaffected by the reporting period.
Automated assessment	The discovery to queue to Nmap to findings pipeline, including how many hosts are vulnerable, safe, or awaiting assessment.
Live monitoring	Traffic and detection as it arrives.

15. Live Monitoring

Live packet capture and traffic analysis, driven through TShark.

STEP 3 — START CAPTURING TRAFFIC

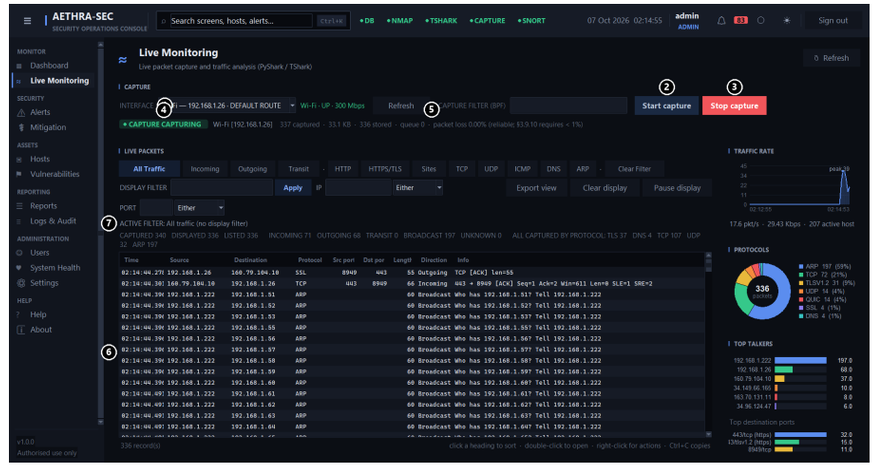


Figure 3. Live Monitoring during a real capture.

①	**Interface selector** The capture adapter. The routing interface is offered first; virtual and disconnected adapters sort after it.
②	**Start capture** Begins capture on the selected interface.
③	**Stop capture** Ends the capture session and its TShark process.
④	**Capture status** Whether capture is running, and on what.
⑤	**Counters** Captured, displayed and dropped, reported separately so the stage under pressure is visible.
⑥	**Packet table** Live packets. Double-click a row to open its full detail.
⑦	**Display filter** Narrows what is shown. It never restarts capture and never starts a second TShark.

Instructions

1. Select **Live Monitoring** in the sidebar.
2. Choose the capture interface. The routing interface is offered first; read Section 10 before choosing another.

3 Leave the capture filter empty to capture everything, or enter a BPF expression to restrict what TShark captures at all.

4 Select **Start capture**. Packets begin arriving within seconds and the status chip changes to CAPTURING.

5 Use the view buttons — All Traffic, Incoming, Outgoing, Transit, HTTP, HTTPS/TLS, DNS, TCP, UDP, ICMP, ARP — to narrow what is displayed. This never restarts capture.

6 Double-click any packet row to open its full detail.

7 Select **Stop capture** when finished. The TShark process ends with it.

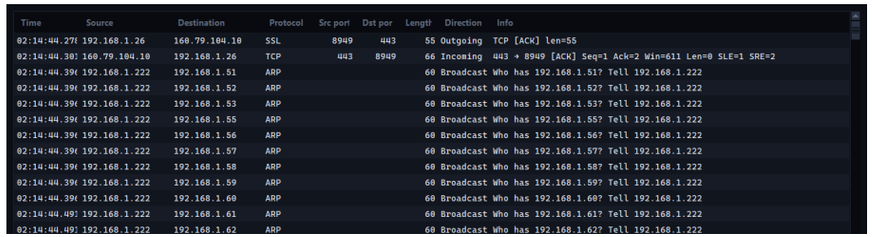


Figure 4. Enlarged: the live packet table. Each row is a real captured frame, with its source, destination, protocol, ports, length and direction.

The capture path

```
network -> Npcap -> TShark -> bounded queue
-> capture worker -> batched UI update / batched writes
```

Capture filter versus display filter

These are separate and behave differently. A **capture filter** is applied by TShark and controls what is captured at all; traffic it excludes is gone. A **display filter** only changes which already captured packets are shown. Changing a display filter never restarts capture and never starts a second capture process.

Views

All, Incoming, Outgoing, Transit, HTTP, HTTPS/TLS, TCP, UDP, ICMP, DNS and ARP. **Transit** means neither endpoint is this machine — traffic between two other devices, shown only when it was genuinely captured.

> **IMPORTANT**
> For HTTPS, AETHRA-SEC shows only metadata the capture actually carries: addresses, ports, TLS version, the server name from a ClientHello when present, and a domain correlated from observed DNS. It does not decrypt traffic and does not extract credentials. Where no domain is available the row reads Unknown/Encrypted rather than showing a guess.

16. IDS Alerts

Alerts received from Snort over the configured transport, alongside anomalies raised by AETHRA-SEC's own traffic rules. The two sources are distinguished rather than merged.

STEP 4 — REVIEW AN ALERT

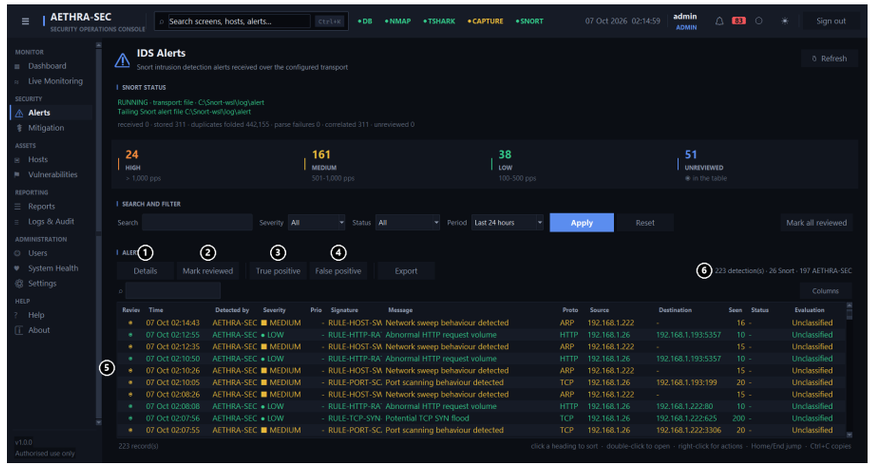


Figure 5. IDS Alerts.

①	**Details** Opens the full alert, including its mitigation recommendation.
②	**Mark reviewed** Records that the alert has been reviewed. The UNREVIEWED marker and the sidebar counter read this same state.
③	**True positive** Records the analyst's evaluation.
④	**False positive** Records the analyst's evaluation.
⑤	**Alert table** Severity, signature, source, destination, protocol, timestamp and review state.
⑥	**Count** How many alerts the current filter matched.

Instructions

1 Select **Alerts** in the sidebar.

2 Select an alert row. Rows not yet reviewed carry the UNREVIEWED marker.

3 Select **Details**, or double-click the row, to open the full alert with its mitigation recommendation.

4 Record your judgement with **True positive** or **False positive**.

5 Select **Mark reviewed**. The UNREVIEWED marker clears and the sidebar counter follows, because both read the same stored state.



Figure 6. Enlarged: the alert table, showing severity, signature, source, destination and review state.

Reviewing an alert

Control	Effect
Details	Opens the full alert, including the mitigation recommendation matched from the knowledge base.
Mark reviewed	Records that the alert has been reviewed. The UNREVIEWED marker and the sidebar counter read the same stored state.
True positive	Records the analyst's evaluation.
False positive	Records the analyst's evaluation.
Export	Exports the currently filtered set, not only the selected row.

NOTE
Acknowledge, Escalate and Benign/expected were removed as unused actions and are not available. Records already in those states keep them and the Status and Evaluation columns still report them truthfully; nothing new can be put into them.

17. Notifications

A notification is created when a real event is recorded — a new alert at or above the configured severity, a finished or failed assessment, or a subsystem failure. The badge count and the alert sound are driven by the same stored record.

Setting	Effect
notifications.enabled	Master switch for creating notifications.
notifications.min_severity	The lowest alert severity that raises a notification.
notifications.sound_enabled	Whether a sound plays.
notifications.sound_min_level	The lowest notification level that sounds.
notifications.sound_dedup_seconds	Repeats of the same finding inside this window fold into the first sound. The record still counts every occurrence.

NOTE
The packaged alert sound is verified at startup. If it is missing or cannot be played, the Settings screen states so and the system sound is used instead — the failure is reported rather than passing as silence.

18. Vulnerability Assessment

Authorised Nmap assessment: discovery, port state, service and OS detection.

STEP 5 — ASSESS A HOST

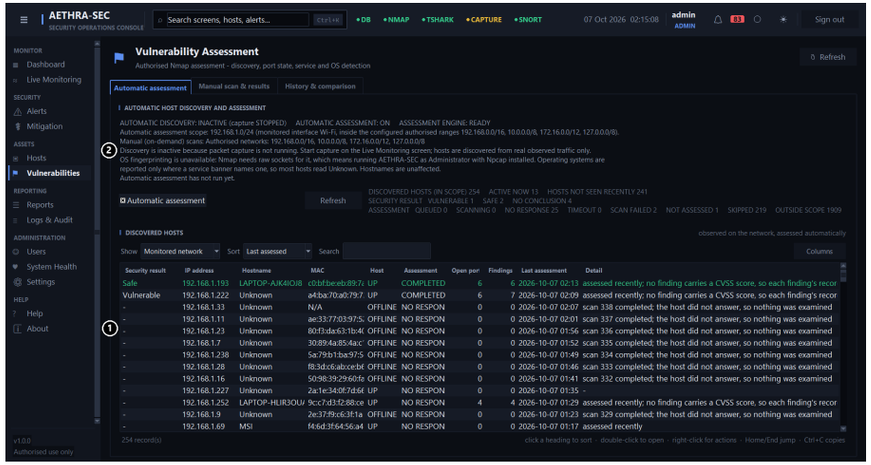


Figure 7. Vulnerability Assessment.

① **Host assessment**
Host, assessment state and security result. Security result is Vulnerable, Safe or a dash only.

② **Automation status**
Whether automatic assessment is running, and what it is working on.

Instructions

1. Select **Vulnerabilities** in the sidebar.
2. Confirm the target is inside the authorised networks. Assessment is refused outside them, in the service layer.
3. Automatic assessment queues discovered hosts on its own. To assess one now, select it and start a scan.
4. Watch the assessment state move through QUEUED and SCANNING to COMPLETED, or to NO RESPONSE if the host never answered.
5. Read the security result: Vulnerable, Safe, or a dash. A dash means no conclusion is available — not that the host is safe.

Three states that are easy to confuse

Dimension	Values	Meaning
Host reachability	UP / OFFLINE / UNKNOWN	Whether the host answered.
Assessment state	NOT ASSESSED / QUEUED / SCANNING / COMPLETED / NO RESPONSE / FAILED	How far assessment has got.
Security result	Vulnerable / Safe / —	What the assessment concluded. A dash means no conclusion is available, not that the host is safe.

IMPORTANT
An open port is exposure, not a vulnerability. A host is reported Vulnerable only on confirmed finding evidence. A rescan supersedes a host's earlier findings by resolving them rather than deleting them, so assessment history is preserved.

19. Hosts

The network as the console currently understands it: hosts observed in captured traffic or discovered by assessment.

STEP 6 — HOST INVENTORY

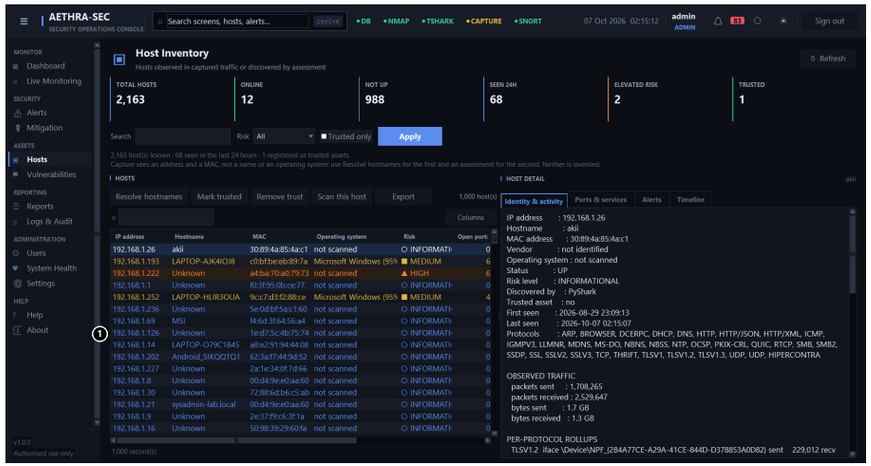


Figure 8. Host Inventory.

① **Host inventory**
Address, hardware address, hostname and its provenance, first and last seen, assessment state and security result.

Instructions

1. Select **Hosts** in the sidebar.
2. Read the hostname together with its provenance: a name learned from an mDNS announcement is not the same evidence as reverse DNS.
3. Check last-seen to judge whether an entry is current.

Column	Source
IP address	Observed in captured traffic or reported by a scan.
MAC address	Recorded only when the frame genuinely carries the host's own hardware address.
Hostname	Reverse DNS, NetBIOS, or a device's own mDNS announcement. The provenance is recorded alongside it.
Operating system	Nmap OS detection, with its confidence.
First / last seen	From real observations.

Column	Source
Assessment status	See Section 18.
Security result	See Section 18.

20. Mitigation

A recommendation-only knowledge base. AETHRA-SEC never applies a mitigation; every recommendation requires a human decision.

STEP 7 — MITIGATION GUIDANCE

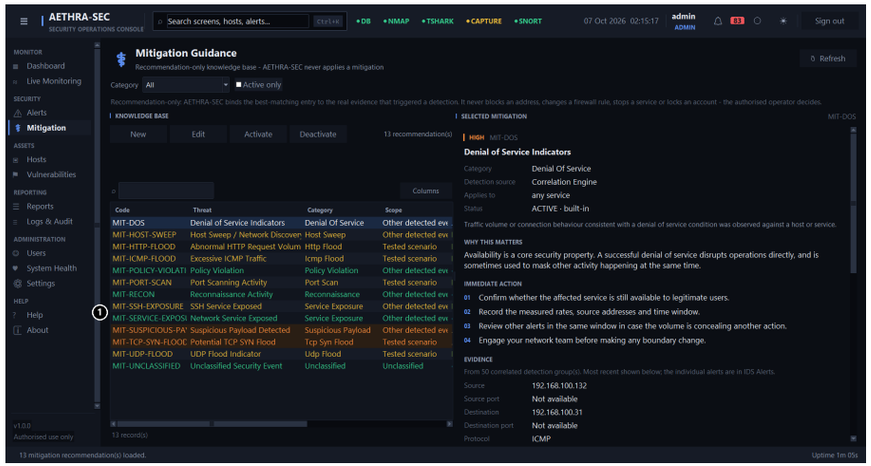


Figure 9. Mitigation Guidance.

① **Knowledge base**
Every recommendation shown against an alert comes from here. AETHRA-SEC never applies one.

Instructions

1. Select **Mitigation** in the sidebar.
2. Find the recommendation matched to a threat category or service.
3. Apply it yourself. AETHRA-SEC never applies a mitigation.

Recommendations shown against an alert are matched from this base by service, severity, threat category or an explicit mapping. When no entry matches, the alert detail says so rather than offering generic advice, and a failed lookup reports the failure.

21. Reports

Reports are generated from stored records only, so every figure is traceable to the rows behind it.

STEP 8 — REPORTS

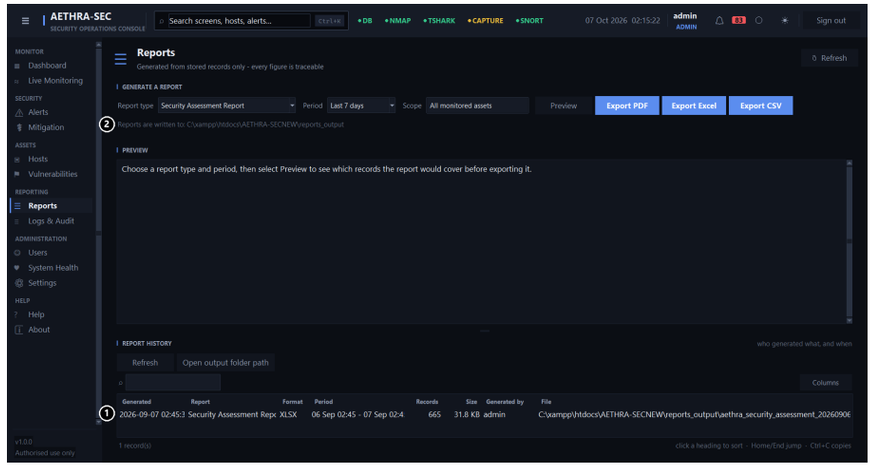


Figure 10. Reports.

① **Report history**
Reports already generated, with scope, period and record counts.

② **Output location**
Where generated reports are written.

Instructions

1. Select **Reports** in the sidebar.
2. Choose a report type, a scope and a period.
3. Generate. The file is written to the reports directory shown on the screen, and the run is recorded so it can be reproduced.

Choose a report type, a scope and a period, then generate. Output is written to the reports directory shown in Section 6, and the generated file, its scope and its record counts are recorded so a report can be reproduced.

22. Logs and Audit

An append-only audit trail with SHA-256 chained integrity hashes: altering a historical entry breaks verification and the application reports it.

STEP 9 — LOGS & AUDIT

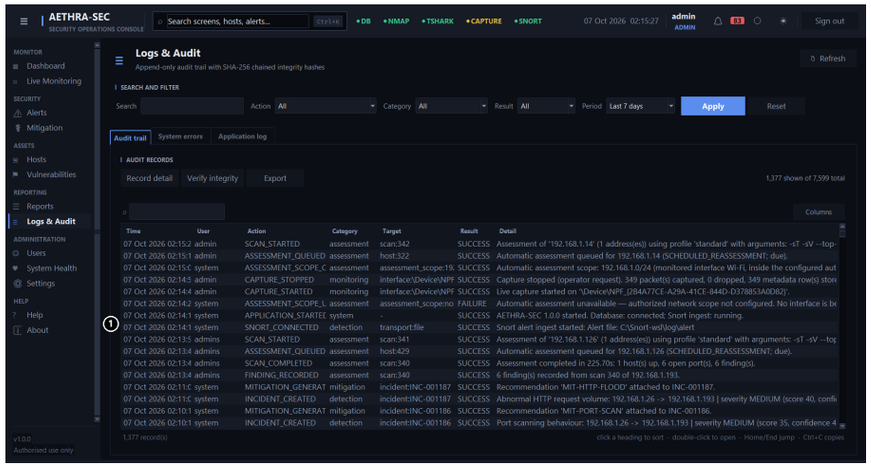


Figure 11. Logs & Audit.

① **Audit trail**
Append-only, with SHA-256 chained integrity hashes.

Instructions

1. Select **Logs & Audit** in the sidebar.
2. Filter by period, actor or action.
3. Export the filtered set if you need it outside the application.

Security-relevant actions are recorded with the actor, the target and the outcome. Sign-in activity, administrative changes and system errors are filterable and exportable. Credentials are never written to the audit trail.

23. User administration

Account registration, password recovery, roles, sign-in history and active sessions. Administrator only.

STEP 10 — USER MANAGEMENT

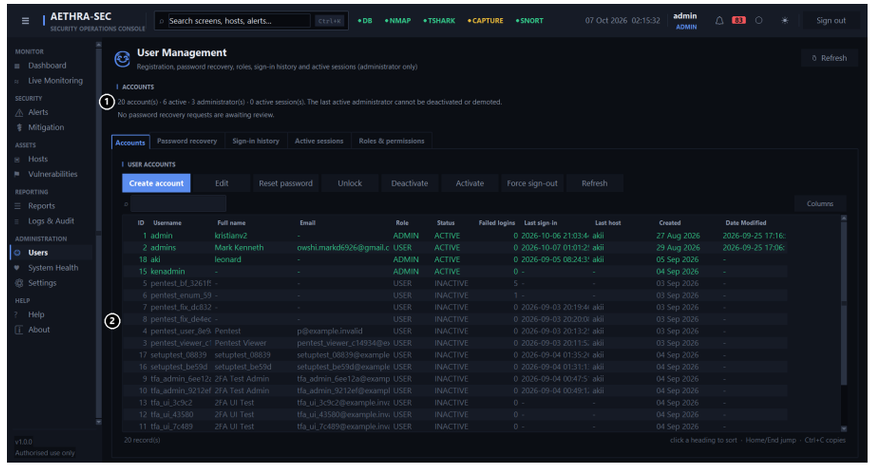


Figure 12. User Management.

① **Account summary**
Totals, administrators and active sessions.

② **Accounts table**
The first column is the database record identifier. Date Modified is the last administrator edit, and stays blank for an account never edited.

Instructions

1. Select **Users** in the sidebar.
2. Select an account, then **Edit** to change its identity or role. Doing so sets Date Modified.
3. Use **Unlock** to clear a lockout, or **Force sign-out** to end an active session.
4. **Create account** adds a new operator. A new account shows a dash under Date Modified until it is first edited.



Figure 13. Enlarged: the accounts table. The first column is the database record identifier; Date Modified is blank for an account never edited.

Column	Meaning
ID	The database record identifier — the real primary key, not a row number.
Username	Sign-in name, unique.
Full name / Email	Display name and contact address.
Role	ADMIN or USER.
Status	Active or inactive.
Failed logins	Consecutive failures; drives lockout.
Last sign-in	Last authentication activity.
Last host	Where that sign-in came from.
Created	Account creation timestamp.
Date Modified	The last time an administrator changed the account's identity or configuration. It does not move on sign-in, on a failed login, or on an automatic lockout. A dash means the account has not been edited since it was created.

Available actions: Create account, Edit, Reset password, Unlock, Deactivate, Activate, Force sign-out and Refresh. The last active administrator cannot be deactivated or demoted.

24. System Health

Live subsystem probes, re-run every time the screen is opened. A component is reported on what was actually observed, never assumed.

STEP 11 — SYSTEM HEALTH

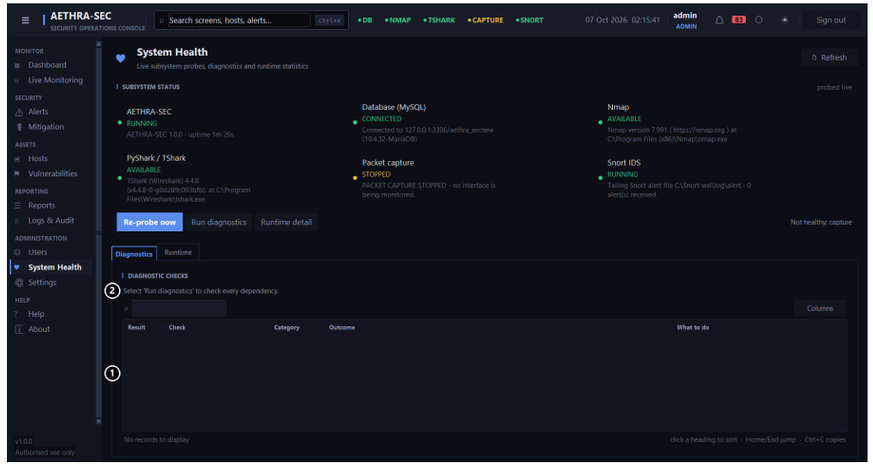


Figure 14. System Health.

① **Subsystem probes**
Database, Nmap, TShark, capture and Snort, re-probed every time this screen is opened.

② **Diagnostics summary**
One line stating whether every component answered.

Instructions

1. Select **System Health** in the sidebar.
2. Every component is re-probed as the screen opens.
3. Read the detail beside any component that is not healthy: it carries the actual error, not a generic message.

Component	What is probed
application	Version, uptime and runtime statistics.
database	Connectivity, schema version and migration state.
nmap	Executable, its path and the version that ran.
tshark	Executable and version; capture availability.
capture	Interface state, counters and drops reported by the capture engine.
snort	Ingest transport state and last alert received.

25. Settings

Seventy-two runtime settings are stored in the database and take effect without restarting, grouped by area. Each carries its purpose, its permitted range and its default in the interface.

STEP 12 — SYSTEM SETTINGS

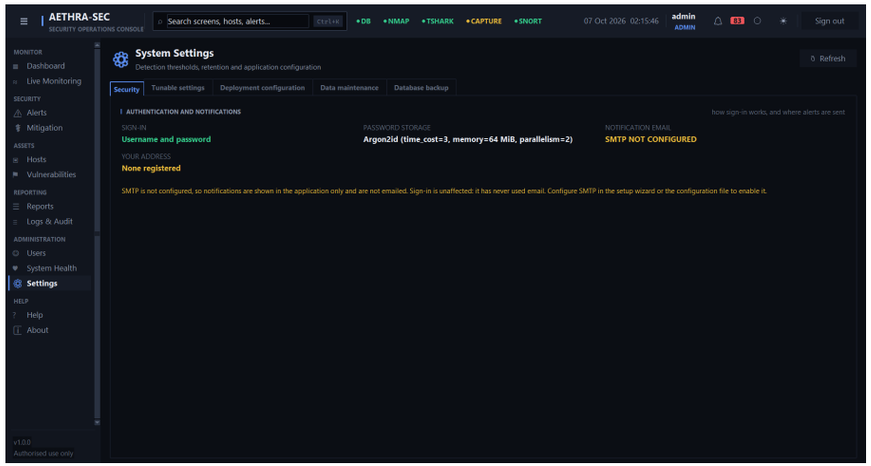


Figure 15. System Settings.

Instructions

1. Select **Settings** in the sidebar.
2. Choose the group you need.
3. Change a value. It is stored in the database and takes effect without restarting.

Group	Count	Covers
detection	13	Anomaly thresholds and detection windows
assessment	11	Automatic assessment scope, cadence, timeouts
capture	11	Queue size, pcap ring bounds, batching, row-rate cap, domain correlation
notifications	8	Enablement, minimum severity, sound
security	8	Lockout, session and password policy
retention	6	How long each class of record is kept
correlation	5	Incident grouping
backup	4	Scheduled backup behaviour
hosts	3	Inventory ageing and offline marking
ui	3	Refresh cadence and table bounds

26. Help and About

STEP 13 — ABOUT AETHRA-SEC

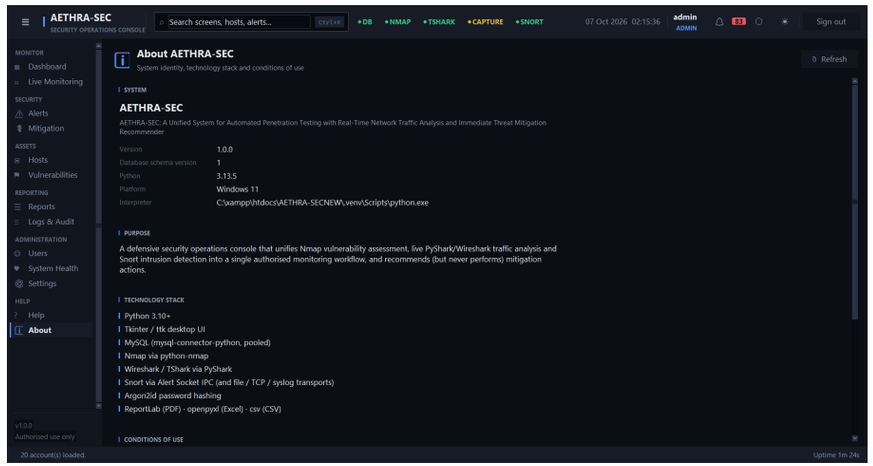


Figure 16. About AETHRA-SEC.

Instructions

1. Select **About** in the sidebar to confirm the version and the conditions of use.

The About screen reports the system identity, version, technology stack and the conditions of use the application operates under. The Help screen covers in-application guidance and keyboard shortcuts, including the command palette.

27. Updating AETHRA-SEC

> **NOTE**
> An automated update procedure is not currently provided by the application. AETHRA-SEC does not check for updates and cannot update itself.

To upgrade, install the newer build over the existing one:

Step	Detail
1. Back up	Take a database backup. The security records live there, not in the install directory.
2. Close	Stop capture and close AETHRA-SEC so no file is in use.
3. Install	Run the newer AETHRA-SEC-Setup.exe. It detects the existing installation and upgrades in place.
4. Migrate	On the next start, migrations newer than the recorded schema version are applied automatically.
5. Verify	Open System Health and confirm the schema version and every component.

> **IMPORTANT**
> Do not drop or recreate the database as part of an upgrade. Migrations are additive and preserve hosts, alerts, findings and the audit trail. Your configuration file is in the data directory and is not replaced by the installer.

28. Uninstalling AETHRA-SEC

Use *Uninstall AETHRA-SEC* from the Start menu, or Apps and Features. The uninstaller is explicit about what it removes.

What	What happens
Application files	Removed from the install directory.
Data directory	**You are asked.** It holds the configuration file, logs, reports and capture files. The default answer is to keep it, and a second confirmation is required to delete it.
Database	**Never removed.** Hosts, alerts, incidents, findings and the audit trail stay in MySQL, because an uninstaller must not destroy security evidence.

Removing the database deliberately

If you genuinely want the security history gone, drop the schema named in the configuration file using your database administration tool. This cannot be undone.

29. Troubleshooting

Symptom	Likely cause and what to check	Resolution
Application will not start	Configuration missing or unreadable; database unreachable at startup.	Run the diagnostics shortcut (--check). Re-run setup with --setup.
Database unavailable	Server not running, wrong host or port, credentials changed.	Start MySQL/MariaDB. Confirm the settings in the configuration file. System Health reports the actual driver error.
Nmap unavailable	Not installed, or not on PATH.	Install Nmap or set its path. Verify with nmap --version.
Npcap unavailable	Not installed, or the console is not elevated.	Install Npcap (included with Wireshark). Start from the shortcut, which prompts for elevation.
TShark unavailable	Wireshark not installed, or TShark not on PATH.	Install Wireshark. Verify with tshark --version.
Snort unavailable	Transport disabled, or no sensor is writing alerts.	Set AETHRA_SNORT_TRANSPORT and confirm the sensor is running. AETHRA-SEC never starts Snort.
No packets captured	Wrong adapter selected; a capture filter is excluding everything.	Confirm the adapter on Live Monitoring is the routing interface. Clear the capture filter.
Other devices not visible	The network is not delivering their traffic to this interface.	Section 10. This needs SPAN, a TAP or a gateway capture point, not a software change.
IDS alerts missing	Snort not writing, transport mismatch, or severity below the notification threshold.	Check System Health, then the alert file path.
Notification sound missing	Sound disabled, level below the threshold, or repeats folded by the dedup window.	Check the notification settings in Section 17. Settings reports whether the packaged sound is usable.
Assessment fails	Target outside the authorised boundary, or Nmap unavailable.	Confirm the target is inside AETHRA_AUTHORIZED_NETWORKS. The failure records the actual Nmap error.
Application feels slow	Very high packet rate, or a large reporting period.	Counters show captured, displayed and dropped separately so the stage under pressure is visible.
Locked out	Consecutive failed sign-ins.	Another administrator unlocks the account from Users.
Report generation fails	Reports directory not writable, or no records in the period.	Check permissions on the reports directory; widen the period.
Installer fails	Not elevated, or files in use.	Run as administrator with AETHRA-SEC closed.

30. Security considerations and limitations

Handling of credentials and secrets

- Account passwords are stored as Argon2id hashes and never in readable form.
- The configuration file holds the database and SMTP passwords. The application tightens its permissions when setup writes it.
- Credentials are never written to the audit trail, the logs or a report.
- Packet detail masks values the dissector identifies as credentials before they reach the interface. The bytes remain in the capture file; only the display is masked.

What AETHRA-SEC cannot do

- It cannot decrypt HTTPS, and does not attempt to.
- It cannot recover Wi-Fi passwords or any other credential.
- It cannot see traffic the network does not deliver to its capture interface, whatever its settings say.
- It cannot start, stop or configure a Snort sensor.
- It does not update itself.

Data retention

Retention settings control how long each class of record is kept. Capture files are bounded by a ring buffer, so a long session cannot fill the disk; a packet whose capture file has been recycled can no longer be inspected in detail, and the inspector says so rather than showing something else.

> **IMPORTANT**
> AETHRA-SEC must only be operated against networks, hosts and interfaces you are explicitly authorised to monitor or assess. It performs no exploitation, no credential attacks and no automated mitigation. All mitigation output is advisory and requires a human decision.

AETHRA-SEC

Security Operations Console

Installation and User Manual

Version 1.0.0

A Unified System for Automated Penetration Testing with Real-Time Network Traffic Analysis and Immediate Threat Mitigation Recommender

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